

MAGNUS DIERKING

Born in Germany, 21 September 1998

email mag.dierking@gmail.com
website magnusdierking.github.io
phone +49 170 9519158

WORK EXPERIENCE

- 08/2025–
Present
IAS Research Assistant
Intelligent Autonomous Systems Group · Technical University of Darmstadt
Developing a Python/C++ software stack for FR3 robotic arms, built on ROS2 Jazzy.
- 10/2024–
05/2025
Huawei R&D Research Intern
Huawei Technologies Research & Development UK Limited · London, Shenzhen
Extracurricular internship during Master studies. Working on reinforcement learning, imitation learning and supporting middleware for robotics as part of the Embodied AI team.
- 09/2023–
09/2024
IAS Research Assistant
Intelligent Autonomous Systems Group · Technical University of Darmstadt
Assisted Ph.D. researchers with experiments, visualization and studies on supporting theory in optimal transport, Bayesian inference and optimization for robot learning.
- 2018–2021,
2022–2023
mas Working Student
Medical Airport Service GmbH · Frankfurt
Support personel for the rescue station on the apron of Frankfurt International Airport.
- 2021–2022
TEMF Teaching Assistant
Institute for Accelerator Science and Electromagnetic Fields · Technical University of Darmstadt
Developed lesson plans, theoretical / programming tutorials and provided mentorship and feedback for undergraduate students.
Courses: Electrodynamics, Numerics for Electromagnetic Field Simulation

PUBLICATIONS

- 2025
arXiv Ark
Open-source, Python-first robotics framework that provides a Gym-style interface for collecting data, training policies, and switching seamlessly between simulation and real-robot deployment. It includes reusable modules for control, SLAM, motion planning and visualization, and integrates natively with ROS to accelerate end-to-end robotics research.
First Author
- 2025
IEEE RA-L OpenPyro-A1
Open-source, low-cost bimanual half-humanoid robot designed for advanced manipulation research. It features a modular, repairable hardware design and supports coordinated two-handed tasks such as folding, cutting, and assembling. The platform enables teleoperation via a Meta Quest 3 and provides interfaces for learning-based controllers to support scalable, real-world robotics experimentation.
Co-Author

EDUCATION

<i>Master of Science</i>	10/2022– <i>Present</i>	Computational Engineering
	GPA: 1.00 · Technical University of Darmstadt Thesis: <i>Model Tensor Planning</i> Ongoing. Advisor: Prof. Jan PETERS · Supervision: Dr. João CARVALHO, Dr. An Thai LE Robot Learning · Reinforcement Learning · Deep Generative Models · Intelligent Robotic Manipulation · Parallel Programming Differential & Riemannian Geometry · Numerical Linear Algebra · Geometric Algebra · Information Theory 1-2 · Convex Optimization · Graph Signal Processing · Control Theory · Optimal Transport	
<i>Bachelor of Science</i>	10/2019– 10/2022	Computational Engineering
	GPA: 1.51 · Top 10% · Technical University of Darmstadt Thesis: <i>Parallel Solution of Linear Systems Arising in Domain Decomposition Methods</i> C++ implementation of a parallel solver for large-scale surface PDEs within an Isogeometric Analysis (IGA) framework. Advisor: Prof. Sebastian SCHÖPS · Supervision: MSc. Maximilian NOLTE Robotics · Software Engineering · Algorithms and Data Structures · Functional & Object-oriented Programming · Geometric Modelling Partial Differential Equations · Numerical and Statistical Methods · Mechanics 1-3 · Signals and Systems · Electrodynamics 1-2 · Maths 1-4	
<i>A-Levels</i>	10/2018– 10/2019	Mechanical Engineering
	<i>Completed Coursework</i> · Technical University of Kaiserslautern Material Science 1-2 · Material Science Lab · Mechanical Design Project · Production Engineering · Business Organization	
	2018	Elisabeth-Langgässer Gymnasium
	Math · English · History	

SKILLS

OS	LINUX · Mostly Ubuntu, some Arch
Programming	PYTHON · JAX, PyTorch, CVX, NumPy, SciPy, Pandas C++ · Eigen, OpenMP Basics · Java, HTML, JavaScript, CSS
Simulation	MuJoCo, PyBULLET
Tools	GIT, ROS2, Docker, L ^A T _E X
Hardware	Franka Research 3, Trossen Viper OptiTrack

OTHER INFORMATION

<i>Scholarships</i>	2024 · Erasmus Placements Program
	2023 · Deutschlandstipendium
<i>Languages</i>	GERMAN · Mothertongue
	ENGLISH · Fluent
	FRENCH · Basics